Data Schema - Nassau Candy Distributor Shipping Analysis

This is the schema to build the Power BI dashboard against. There are two versions here: the flat table the project actually uses (simple, fine for a dataset this size), and a star schema version showing how I'd model it if this were a bigger, ongoing production dataset. Bring the flat table into Power BI unless you specifically want to practice building relationships between tables, in which case use the star schema section below.

Flat table (what to actually import)

Table name: `shipments`

Source file: `Nassau_Candy_Distributor_clean.csv`, the output of `data_cleaning.py`.

Column	Type	Notes
Row_ID	Whole Number	
Order_ID	Text	

Column	Type	Notes
Order_Date	Date	parsed from DD-MM-YYYY
Ship_Date	Date	unreliable at row level, see note below
Ship_Mode	Text	Standard Class / Second Class / First Class / Same Day
Customer_ID	Whole Number	
Country_Region	Text	United States / Canada
City	Text	
State_Province	Text	set Data Category to "State or Province" in Power BI so the map visual works
Postal_Code	Text	

Column	Type	Notes
Division	Text	Chocolate / Sugar / Other
Region	Text	Pacific / Atlantic / Interior / Gulf
Product_ID	Text	
Product_Name	Text	
Sales	Decimal	
Units	Whole Number	
Gross_Profit	Decimal	
Cost	Decimal	
Shipping_Lead_Time_Days	Decimal	derived from Ship_Mode, not the raw Ship_Date column
Factory	Text	mapped from Product_Name
Route	Text	Factory + " to " +

Column	Type	Notes
		State_Province
Order_Month	Text	YYYY-MM, for the trend chart

Note on Ship_Date: don't build any measure directly off this column. Every row in the source data is off by 900+ days from Order_Date, so it's not usable for calculating a real lead time.
Shipping_Lead_Time_Days is the column to use instead, it's a Ship Mode based estimate built during the cleaning step.

Star schema (for a larger or production setup)

If this dataset were bigger, or being refreshed daily, or shared across a team, I'd split it into a fact table and five dimension tables instead of one flat table.

Fact_Shipments

- Order_ID
- Order_Date
- Ship_Mode_ID (foreign key)
- State_ID (foreign key)
- Factory_ID (foreign key)

- Product_ID (foreign key)
- Sales
- Units
- Gross_Profit
- Cost
- Shipping_Lead_Time_Days

Dim_Date

- Date
- Year
- Month
- Quarter
- Order_Month

Dim_ShipMode

- Ship_Mode_ID
- Ship_Mode

Dim_Geography

- State_ID
- State
- Region
- Country

Dim_Factory

- Factory_ID

- Factory
- Latitude
- Longitude

Dim_Product

- Product_ID
- Product_Name
- Division

Why I didn't build it this way for this project: one analyst, one dataset, fast iteration, and normalizing it would have added modeling overhead without a real payoff at this scale. It's worth knowing the star schema version exists and being able to explain when it's the better call (bigger data, multiple people querying it, needing reusable dimensions), which is usually a question that comes up in interviews.

How to set this up in Power BI specifically

1. Get Data, choose Text/CSV, point it at the cleaned CSV.
2. Go to the Data pane and check that Order_Date came in as a Date type, not Text. Fix it in Power Query if it didn't.
3. Click State_Province, go to Column Tools, Data Category, and set it to "State or Province". This is

the step people skip and then wonder why their map visual won't render.

4. Create the DAX measures listed in `powerbi_dashboard_spec.md`.
5. Build the visuals following the layout in that same file.